

UNIVERSITY OF KENTUCKY: COLLEGE OF MEDICINE

Institute for Biomedical Informatics

BMI 730: Advanced Artificial Intelligence in Clinical Practice and Systems Implementation

Term: Fall 2026

Credit Hours: 3.0

Format: Hybrid (Canvas Asynchronous Modules + Synchronous Clinical Seminars)

Location: UK Albert B. Chandler Hospital, Pavilion A, Room [TBA] & Zoom

Part I: Instructor and Course Information

Instructor Contact

- **Instructor:** [Your Name, MD / Title]
- **Office Location:** Pediatric Emergency Department / IBI Faculty Offices
- **Office Hours:** Wednesdays 10:00 AM – 12:00 PM (In-person or via Teams)
- **Email:** [Your Email]@uky.edu
- **Communication Policy:** Due to clinical shift schedules, asynchronous communication via Canvas Inbox is required for all non-emergent course matters. Expect a response within 24–48 hours. For time-sensitive issues regarding API access or Canvas submissions, prefix your subject line with [URGENT: BMI 730].

Course Description

This advanced graduate and medical elective explores the direct integration of Artificial Intelligence (AI) and Large Language Models (LLMs) into high-stakes clinical workflows and healthcare logistics. Moving beyond introductory AI literacy, this course treats AI as an active clinical and operational tool. Students will critically evaluate LLM performance in medical education, design autonomous logistics frameworks for rural healthcare delivery, and analyze the efficacy of predictive modeling in emergency and pediatric medicine. The course culminates in a systems-level architecture proposal aligned with the UK Artificial Intelligence Task Force guidelines.

Student Learning Outcomes (SLOs)

Upon successful completion of this course, students will be able to:

1. **Evaluate** the diagnostic and evaluative accuracy of foundation models (e.g., ChatGPT-4o, Claude) in grading complex clinical documentation, such as standardized patient SOAP notes.
2. **Architect** secure, HIPAA-compliant workflows for deploying generative AI within electronic health record (EHR) environments.
3. **Design** operational frameworks for AI-driven healthcare logistics, specifically addressing challenges in rural trauma care and autonomous delivery networks.
4. **Critically Analyze** the ethical, legal, and systemic risks of predictive algorithms in emergency medicine, with a focus on pediatric populations and vulnerable demographics.
5. **Formulate** institutional governance strategies that align with current state and enterprise-level task force initiatives (e.g., AI@UK).

Required Materials and Technologies

- **Primary Platform:** Canvas for all modules, rubric distribution, and assignment submissions.
- **Software:** Microsoft Teams (for synchronous seminars), Visual Studio Code (for reviewing Python/API integration examples).
- **Textbooks & Literature:** * Topol, E. (2019). *Deep Medicine: How Artificial Intelligence Can Make Healthcare Human Again*.
 - Weekly primary literature, including quantitative studies on LLM performance and systemic reviews of autonomous medical logistics (hosted natively via Canvas).
- **Access:** Students will be provided controlled API access to designated LLMs via a secure, university-provisioned sandbox for data-sensitive assignments.

Part II: Grading and Evaluation

Grade Breakdown

This course utilizes a rigid, points-based grading system to ensure objective evaluation of highly subjective technical architectures.

Assessment Category	Points	Percentage of Final Grade
Clinical Literature Synthesis (4 x 25 pts)	100	20%
HIPAA & AI Compliance Certification Exam	50	10%
Hands-On Clinical AI Simulations (2 x 75 pts)	150	30%
Midterm: Systems Architecture Blueprint	75	15%
Final Applied Integration Project & Presentation	125	25%
Total Possible Points	500	100%

Graduate Grading Scale (UK Senate Policy for 700-level courses):

- **A:** 450 – 500 points (90 – 100%)
- **B:** 400 – 449 points (80 – 89%)
- **C:** 350 – 399 points (70 – 79%)
- **E:** Below 350 points (< 70%)

(Note: 'D' grades are not awarded at the graduate level).

Part III: Strict Course Policies

Generative AI Acceptable Use & HIPAA Compliance Policy

As a course focused on clinical AI, you will heavily utilize generative models. However, strict boundaries apply:

- **PHI Prohibition:** Under NO CIRCUMSTANCES are you to input real Protected Health Information (PHI) or real patient scenarios into public, consumer-facing AI models (e.g., standard ChatGPT, free Claude). Doing so is a federal HIPAA violation, grounds for immediate failure of the course, and will be reported to the UK College of Medicine Dean's Office.
- **De-Identified/Synthetic Data:** All assignments must use the synthetic, de-identified patient datasets provided in Canvas.
- **Transparency Requirement:** Any architecture document, codebase, or literature synthesis that utilizes AI for drafting or coding must include a **Model Usage**

Appendix detailing the model used, the prompt chain, and the validation methods employed to prevent clinical hallucinations.

Academic Integrity

Per Senate Rule 6.3.1, academic dishonesty, including the falsification of data or the uncredited use of AI to generate core intellectual synthesis, is strictly prohibited. You are expected to be the primary intellectual driver of your work.

Accommodations Due to Disability

If you have a documented disability requiring academic accommodations, provide a Letter of Accommodation from the Disability Resource Center (drc@uky.edu) by the end of Week 2 to ensure timely implementation of your plan.

Part IV: Granular Weekly Schedule

All asynchronous Canvas modules unlock on Mondays at 8:00 AM EST. Assignments are due Fridays at 11:59 PM EST unless a synchronous seminar is noted.

Module 1: The Clinical AI Landscape

Week 1 (Aug 24 – 28): Introduction to Clinical Informatics and Foundation Models

- **Focus:** Differentiating between standard machine learning (predictive analytics) and generative AI in the clinical space.
- **Readings:** Topol, Chapters 1-2; UK Healthcare AI Governance Framework (Canvas).
- **Deliverable:** Canvas Discussion - Baseline assessment of current clinical pain points.

Week 2 (Aug 31 – Sep 4): HIPAA, Data Privacy, and Secure Enclaves

- **Focus:** The mechanics of data de-identification, secure APIs, and the legal landscape of PHI processing by third-party compute.
- **Readings:** "The Privacy Calculus of Clinical AI" (Canvas PDF).
- **Deliverable:** **HIPAA & AI Compliance Certification Exam (Due Sep 4).** *Must pass with 100% to proceed to hands-on simulations.*

Module 2: AI in Medical Education and Documentation

Week 3 (Sep 7 – 11): The Ambient Scribe

- *No class Monday, Sep 7 (Labor Day).*
- **Focus:** Natural Language Processing (NLP) for real-time clinical documentation. Analyzing current market leaders (e.g., Nuance DAX).
- **Readings:** Selected literature on physician burnout and documentation burdens.
- **Deliverable:** Clinical Literature Synthesis 1.

Week 4 (Sep 14 – 18): Evaluating LLMs in Medical Education

- **Focus:** Quantitative evaluation of AI models acting as clinical educators.
- **Simulation 1:** Students will use a provided dataset of 50 synthetic medical student SOAP notes. You will design a prompt architecture for an LLM to accurately grade these notes against a standard rubric, comparing the AI's variance to human attending physicians.
- **Deliverable: Simulation 1 Report: SOAP Note Evaluation Architecture (Due Sep 18).**

Module 3: Predictive Analytics & Emergency Medicine

Week 5 (Sep 21 – 25): Predictive Triage and Algorithmic Bias

- **Focus:** Implementing predictive models in the Emergency Department (ED). Analyzing the risks of reinforcing systemic biases in triage scoring.
- **Readings:** "Algorithmic Bias in Clinical Decision Making" (Canvas).
- **Deliverable:** Clinical Literature Synthesis 2.

Week 6 (Sep 28 – Oct 2): Pediatric Considerations in Clinical AI

- **Focus:** Why models trained primarily on adult data fail in pediatric emergency medicine. Adjusting for developmental variances and physiological baselines.
- **Deliverable:** Canvas Discussion - Designing safe fail-states for pediatric predictive models.

Week 7 (Oct 5 – 9): Midterm Evaluation

- **Deliverable: Midterm: Systems Architecture Blueprint (Due Oct 9).** Draft a comprehensive proposal for integrating a specific AI tool into a hospital department, including network topology, API routing, and user workflow.

Module 4: Autonomous Systems and Rural Healthcare Logistics

Week 8 (Oct 12 – 16): AI Beyond the Screen: Medical Logistics

- **Focus:** The intersection of AI routing algorithms and physical healthcare delivery in resource-limited environments.
- **Readings:** Topol, Chapter 10; Review of FAA regulations regarding autonomous flight.
- **Deliverable:** Clinical Literature Synthesis 3.

Week 9 (Oct 19 – 23): Case Study: Drone Networks for Trauma Resuscitation

- **Focus:** Systems engineering for rural medicine.
- **Simulation 2:** Designing the logic parameters for a drone delivery network tasked with transporting whole blood to rural trauma scenes in Appalachia. Students will model variables such as payload weight, temperature control telemetry, weather routing algorithms, and dispatch integration with 911 centers.
- **Deliverable: Simulation 2 Report: Rural Drone Logistics Parameters (Due Oct 23).**

Week 10 (Oct 26 – 30): Fall Break & Integration

- *No classes Oct 26–27 (Fall Break).*
- **Focus:** Independent research week to finalize the scope of the Final Applied Project.

Module 5: Institutional Governance and Implementation

Week 11 (Nov 2 – 6): AI at the Enterprise Level

- **Focus:** Scaling pilot projects to hospital-wide initiatives. Navigating C-suite approvals, IT security reviews, and physician adoption.
- **Readings:** Case studies from the UK Artificial Intelligence Task Force.
- **Deliverable:** Clinical Literature Synthesis 4.

Week 12 (Nov 9 – 13): Change Management and Faculty Training

- **Focus:** How to train clinical staff to recognize AI hallucinations and use models as "co-pilots" rather than absolute authorities.
- **Deliverable:** Canvas Discussion - Drafting a one-page "AI Quick Start" guide for attending physicians.

Week 13 (Nov 16 – 20): Final Project Prep & Peer Review

- **Focus:** Synchronous Zoom seminars for peer review of Final Project methodologies.
- **Deliverable:** Draft of Final Project Methodology Appendix.

Week 14 (Nov 23 – 27): Thanksgiving Break

- *No classes Nov 25–28.*

Week 15 (Nov 30 – Dec 4): Project Consultations

- **Focus:** Open office hours. Dedicated time for debugging code, refining prompt architectures, and perfecting presentations.

Week 16 (Dec 7 – 11): Final Submissions

- **Deliverable: Final Applied Integration Project Due (Dec 9 at 11:59 PM).** This includes a detailed architecture document, the accompanying codebase or prompt templates, and an executive summary formatted for an institutional review board.

Finals Week (Dec 14 – 17)

- **Deliverable:** Synchronous Final Presentations (Schedule TBA). Students will pitch their Final Applied Project to a panel of peers and clinical informatics faculty.

Would you like me to draft the exact prompt architecture parameters and grading rubric for the **Week 4 Simulation (Evaluating LLMs on Medical Student SOAP Notes)**, or would you prefer I outline the technical requirements for the **Week 9 Rural Drone Logistics** assignment?